https://xtremepape.rs/

Question	Answer	Marks
1	**Defining the problem**	
	L is the independent variable and *s* is the dependent variable **or** vary *L* and measure *s*	**1**
	keep mass of load or *M* constant	**1**
	Methods of data collection	
	labelled diagram of workable experiment including: • method of fixing strip at one end, e.g. with a G-clamp or (heavy) mass placed on top of strip over bench • load shown touching at point P • load and G-clamp or (heavy) mass labelled	**1**
	method of attaching load to strip, e.g. use glue/tape/attach with a hook and string	**1**
	use a rule to measure *L* and *s*	**1**
	use a balance to measure *M*	**1**
	Method of analysis	
	plot a graph of *s* against L^3 (allow lg *s* against lg *L*, or log)	**1**
	relationship valid if a straight line through (0,0) (for lg *s* against lg *L* gradient of straight line = 3)	**1**
	$E = \frac{4Mg}{bt^3 \times \text{gradient}}$ (for lg *s* against lg *L*, $E = \frac{4Mg}{bt^3} \div 10^{y-\text{intercept}}$)	**1**

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Question	Answer	Marks
1	**Additional detail including safety considerations**	**Max. 6**
	D1 use cushion/foam/sandbox in case mass/load falls **or** wear goggles in case strip snaps or recoils	
	D2 use same wooden strip or keep *b* and *t* constant	
	D3 clamp rule vertically to measure *s*	
	D4 method to ensure clamped rule to measure *s* is vertical, e.g. correctly positioned set square indicated at right angles to the horizontal surface or plumb line shown in appropriate position	
	D5 *s* = reading of vertical rule with loaded strip – reading of vertical rule with no load	
	D6 repeat *s* measurement for each *L* (unloading and loading strip) and average *s*	
	D7 use a micrometer/calipers to determine *t*	
	D8 repeat readings for *b* and/or *t* at different points along/across the strip and average	
	D9 method to ensure strip is perpendicular to the bench, e.g. repeat measurements of *L* on each side of strip to check that *L* is constant or set square correctly indicated on diagram	
	D10 wait until block is stationary/in equilibrium or measure *s* after a fixed time	

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